ENGINEERING PLAN · MICROSOFT 365 EDITION · OCTOBER 2026

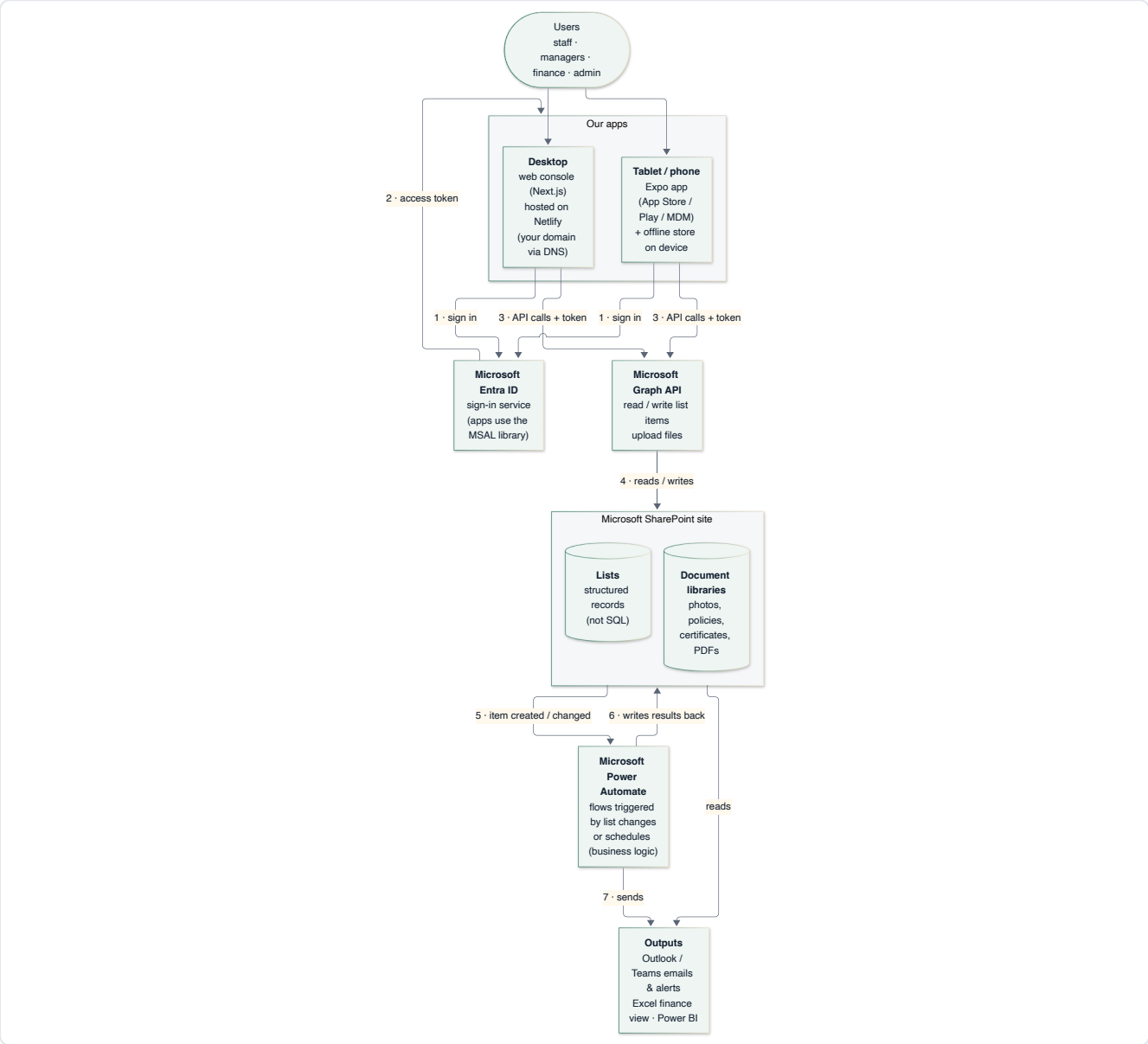
Tech stack and architecture plan

Our own tablet, phone and web apps on top of Maginhawa's Microsoft 365: SharePoint for data and files, Power Automate for business logic, Entra ID for sign-in. Includes the SharePoint list design (31 lists, 7 libraries), the flow catalogue, the EOD trading module, security, scalability and limits, licensing and the build order.

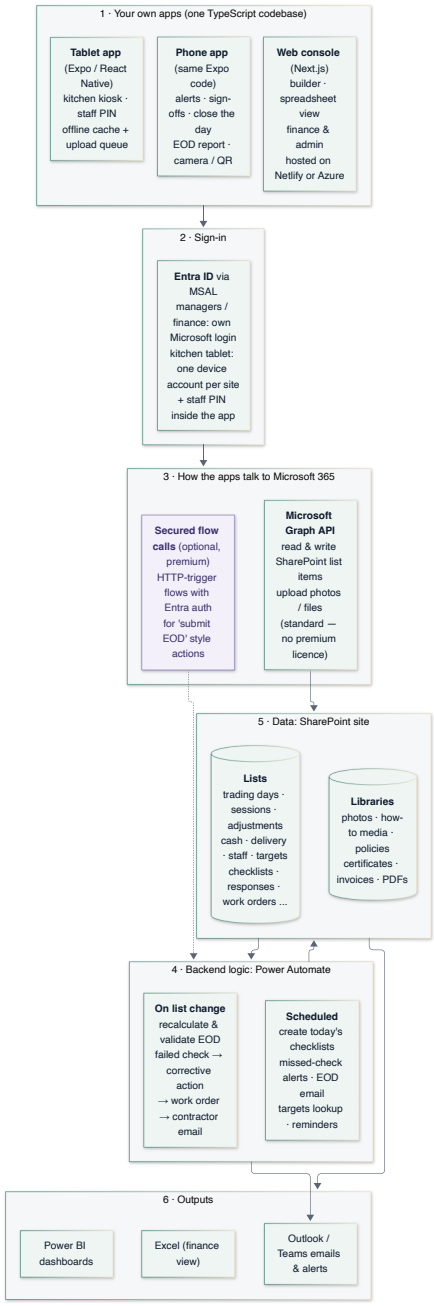
> **Decision:** we build **our own apps** (a kitchen tablet app, a phone app and a web console) and keep Maginhawa's **Microsoft 365** as the backend: **SharePoint** stores the data and files, **Power Automate** runs the business logic, and **Entra ID** handles sign-in. There is no separate server or database to run. Users get fast, purpose-built apps that work offline. Finance keeps Excel, Outlook and Power BI. Running costs stay inside the existing Microsoft 365 licences as long as we stick to **standard connectors**. This setup comfortably handles Maginhawa's **9 sites**. With the design rules in the scalability section it stretches to roughly 20–40 sites. It is an **internal tool**. If the product is later sold to other restaurants, or grows past ~50 sites, the data moves to a database. The Postgres design for that is kept in the `postgres-option/` folder, and keeping all data access in one module makes the move a backend swap rather than a rewrite. Several regulatory values, such as temperature limits and inspection intervals, are still **[verify]** items from the product plan. They live as data in SharePoint, never in code.

The simple picture

Users open **our apps**. The apps sign in through **Microsoft Entra ID** (using the MSAL library), then use the token to call the **Microsoft Graph API**, which reads and writes the **SharePoint** lists and libraries. **Power Automate** sits *beside* SharePoint, not inside Graph: flows start when items change or on a schedule, write results back to SharePoint, and send emails and alerts. Excel and Power BI read SharePoint for finance.



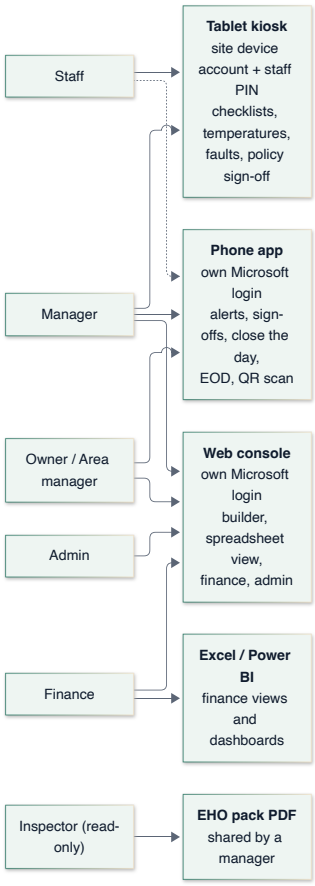
Architecture at a glance



Layer	Choice	Why	Notes
Tablet + phone app	**Expo (React Native)**, TypeScript	One codebase for iPad and Android tablets and phones. Native speed, an offline store and the camera. Over-the-air updates for quick fixes	The tablet layout is a shared kiosk; the phone layout is for managers
Web console	**Next.js** (React) + Tailwind; a data-grid component for the spreadsheet view; ExcelJS export	Builder, spreadsheet view, finance and admin screens. Shares types and the data module with the mobile app	Hosted on **Netlify** (already used) or Azure Static Web Apps
Sign-in	**Entra ID** via **MSAL**	Managers, finance and admins use their own Microsoft accounts. Each kitchen tablet uses a **site device account**, and staff identify themselves with a **PIN** inside the app	Confirm device-account licensing with Maginhawa's Microsoft partner
Data access	**Microsoft Graph API** (SharePoint list items and drive files)	A standard API, so **no premium licence** and no public URLs. Supports `$batch` (20 requests per call) and `Retry-After` throttling	Wrapped in one `data` module in our code (rule 9)
Data	**SharePoint site "Maginhawa Ops"**: 31 lists + 7 document libraries	Structured records in lists; photos, policies, certificates, invoices and PDFs in libraries	Version history on; indexes and unique keys defined below

Layer	Choice	Why	Notes
Business logic	**Power Automate**: 11 flows, owned by a **service account**	Calculations, checks, alerts, emails, scheduling and escalation	Triggered by list changes or on a schedule, not by public HTTP calls
Finance views	**Excel** (Power Query reading the lists) + **Power BI**	Finance keeps working in Excel; dashboards for sales vs target, wage %, missed checks and repairs	Power BI Pro licences for viewers (check the existing licence)
Notifications	**Outlook** and **Teams** connectors (standard)	EOD emails, missed-check alerts, reminders and contractor emails	SMS via Twilio is premium, so it comes later
Device management	**Intune** (if licensed): iPad Guided Access / Android kiosk mode	Locks the shared tablet to our app	
Later, optional	**Azure Function "AI proxy"** + Claude API; EPOS, rota and delivery connectors; Xero/QuickBooks/Sage	The first small piece of our own backend, needed only because API keys can't live in the apps	Stage 3

Who uses what

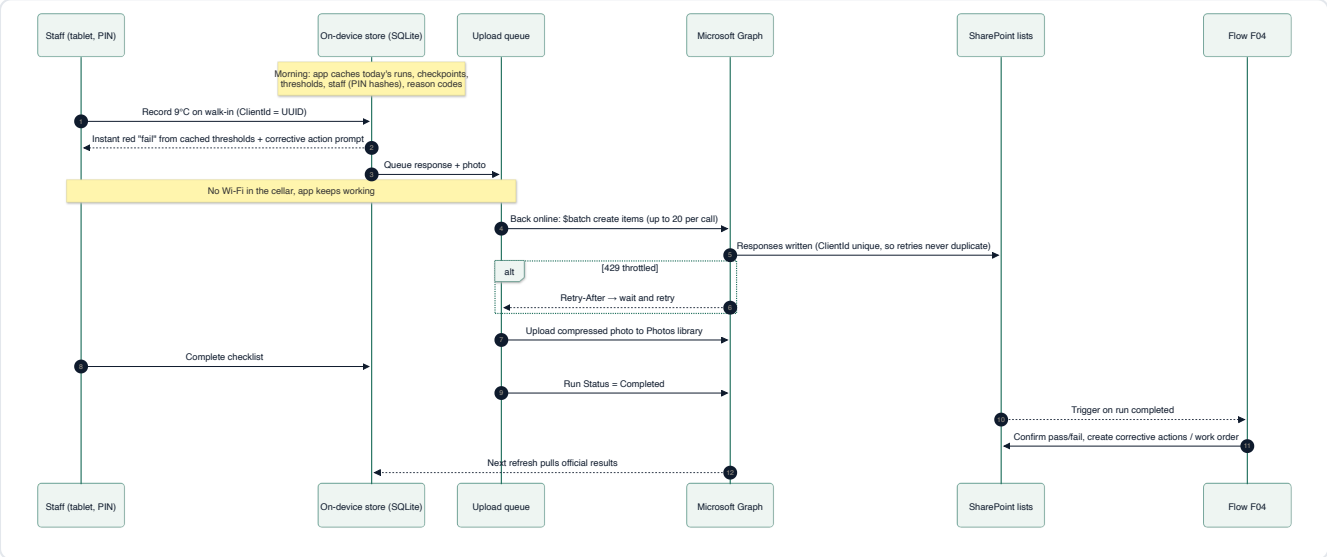


- **Kitchen tablet.** It is signed in with that site's **device account**. Staff tap their name and enter a 4-digit PIN. The PIN is checked against an argon2 hash from the `Staff` list that is cached on the device, so login works offline. The session auto-locks after inactivity. Every record stores the staff member, the device and both timestamps.
- **Phone app.** Each manager or owner signs in with their own Microsoft account. They get alerts for exceptions only, sign off corrective actions, close the day, submit the EOD and scan asset QR codes. On small sites without a tablet, staff can use phones too.
- **Web console.** Admins, managers, finance and owners use it to build checklists, view the spreadsheet summary, manage finance, users and settings, and generate the EHO pack.
- **Roles.** Roles come from Entra **security groups** (Staff, Managers, Finance, Admins/Owners) plus the `AppRole` column on `Staff`. Kiosk-only staff need no Microsoft licence.

Offline-first in our apps

Kitchens and cellars have poor signal. The apps never wait on the network:

1. **Morning cache.** On opening, the app downloads today's checklist runs, checkpoints, thresholds, staff, reason codes and assets into an on-device SQLite store.
2. **Local first.** Every answer saves on the device instantly. The app shows pass or fail straight away from the cached thresholds and prompts for a corrective action on failure.
3. **Upload queue.** Writes are queued and sent through Graph `$batch` when the connection returns. Each record carries a **`ClientId` (UUID)** stored in a unique column, so retries never create duplicates.
4. **Photos.** Photos are compressed on the device (~300 KB) and uploaded in the background to the `Photos` library.
5. **Official result.** Power Automate re-checks the results afterwards (flow F04 for checklists, F01 for the EOD) and writes the official values. The app refreshes and shows them.
6. **Throttling.** When Microsoft throttles requests (HTTP 429), the queue waits for the `Retry-After` time and tries again. Nothing is lost.



SharePoint site design

One SharePoint site, **"Maginhawa Ops"**, holds everything. Lists store structured records, and libraries store files. Each table in the Postgres design maps to a list, so moving to a database later is a straight copy. The full column-by-column specification is in **Appendix A** and in the machine-readable file `sharepoint/sharepoint-lists.json`, which will also drive the provisioning script.

Group	List	Purpose	Written by	Unique key
Reference	**Sites**	One row per restaurant; drives rules, targets and settings	Admin (console)	SiteCode
Reference	**Staff**	Everyone who uses the apps, incl. kiosk-only staff without Microsoft accounts	Admin / Manager	–
Reference	**Devices**	Registered kiosk tablets and phones	Admin	–
Reference	**Targets**	Daily targets per site (imported from finance Excel or edited in console)	Flow F09 / Finance	SiteCode|yyyy-mm-dd
Reference	**ReasonCodes**	Discount / comp / void reasons (23 discount/comp + 15 void codes from the current form)	Finance / Admin	–
Reference	**DeliveryPlatforms**	Deliveroo, Uber Eats, Just Eat…	Finance	Platform name

Group	List	Purpose	Written by	Unique key
Reference	**RulesThresholds**	Food-safety limits per nation (data, not code); consultant-verified	Admin (consultant sign-off)	Nation|CheckCode
Checklists	**ChecklistTemplates**	Checklist definitions (opening, closing, cleaning, food safety, custom)	Admin / Manager (console builder)	–
Checklists	**Checkpoints**	Questions inside a template version — the customisable checkpoints with media	Admin / Manager (builder)	–
Checklists	**Schedules**	When each checklist is due at each site	Admin / Manager	–
Checklists	**ChecklistRuns**	One instance of a checklist on a day (created each morning by flow F02)	Flow F02 creates; app completes	SiteCode|yyyy-mm-dd|ScheduleId
Checklists	**Responses**	Every answer — the compliance record (largest list)	App (append-only by convention + Purview record label)	ClientId
Checklists	**CorrectiveActions**	What was done about a failed check	App + flows F04/F05	–
Checklists	**DayClosures**	Ops end-of-day sign-off per site	App + flow F01	SiteCode|yyyy-mm-dd
Checklists	**HandoverNotes**	Shift handover log	App	–
Maintenance	**Assets**	Equipment register with QR sticker per unit	Manager / Admin	QRCode
Maintenance	**Contractors**	Refrigeration, gas, electrical, extraction, pest…	Manager / Admin	–
Maintenance	**WorkOrders**	Repairs: from fault report, failed check, statutory due or EOD note	App + flows F05/F06	Reference WO-2026-00123
Maintenance	**ComplianceItems**	Statutory calendar: gas CP42, TR19, EICR, fire alarm, emergency lighting, LOLER, PSSR, legionella, pest	Admin / flow F07	–
Maintenance	**Invoices**	Contractor invoices for the Finance role	Finance	–
Maintenance	**VisitorLog**	Contractor and visitor sign-in	App	–
Trading (EOD)	**TradingDays**	One per site per service date — replaces the Netlify form's Excel row	App creates; flow F01 writes official totals	SiteCode|yyyy-mm-dd
Trading (EOD)	**TradingSessions**	Lunch / dinner figures and cash	App; flow F01 adds net & variance	SiteCode|yyyy-mm-dd|Session
Trading (EOD)	**Adjustments**	Discounts, comps and voids as rows (were text blobs)	App	ClientId (UUID)
Trading (EOD)	**DeliverySales**	Per platform per day	App; flow F01 adds estimated net	SiteCode|yyyy-mm-dd|Platform
Trading (EOD)	**DeliveryRefunds**	Refund requests from delivery platforms	App	–
Trading (EOD)	**ShiftStaff**	Who worked each session (fixes lunch managers being dropped)	App (rota later)	–
Trading (EOD)	**Feedback**	Service, kitchen, staff, complaints, maintenance notes	App; flow F01 links work orders	–
People & handbook	**PolicyAcknowledgements**	E-signed policy sign-offs	App	PolicyFileId|Version|StaffId
People & handbook	**StaffCertificates**	Level 2/3 food hygiene, personal licence, first aid…	Manager	–
People & handbook	**FitnessToWork**	Illness exclusion / return to work — restricted to Owner/Admin	Manager (restricted list)	–

Document libraries

Library	Holds	Notes
Photos	Check, fault and work-order photos	Folders Site/yyyy/mm; images compressed on device (~300 KB)
HowToMedia	Images and short videos attached to checkpoints	Read-only for staff
Policies	Handbook and policies (Word/PDF) with metadata: Category, RequiresAck, AppliesToRoles	Major versions only; publishing a major version triggers flow F08

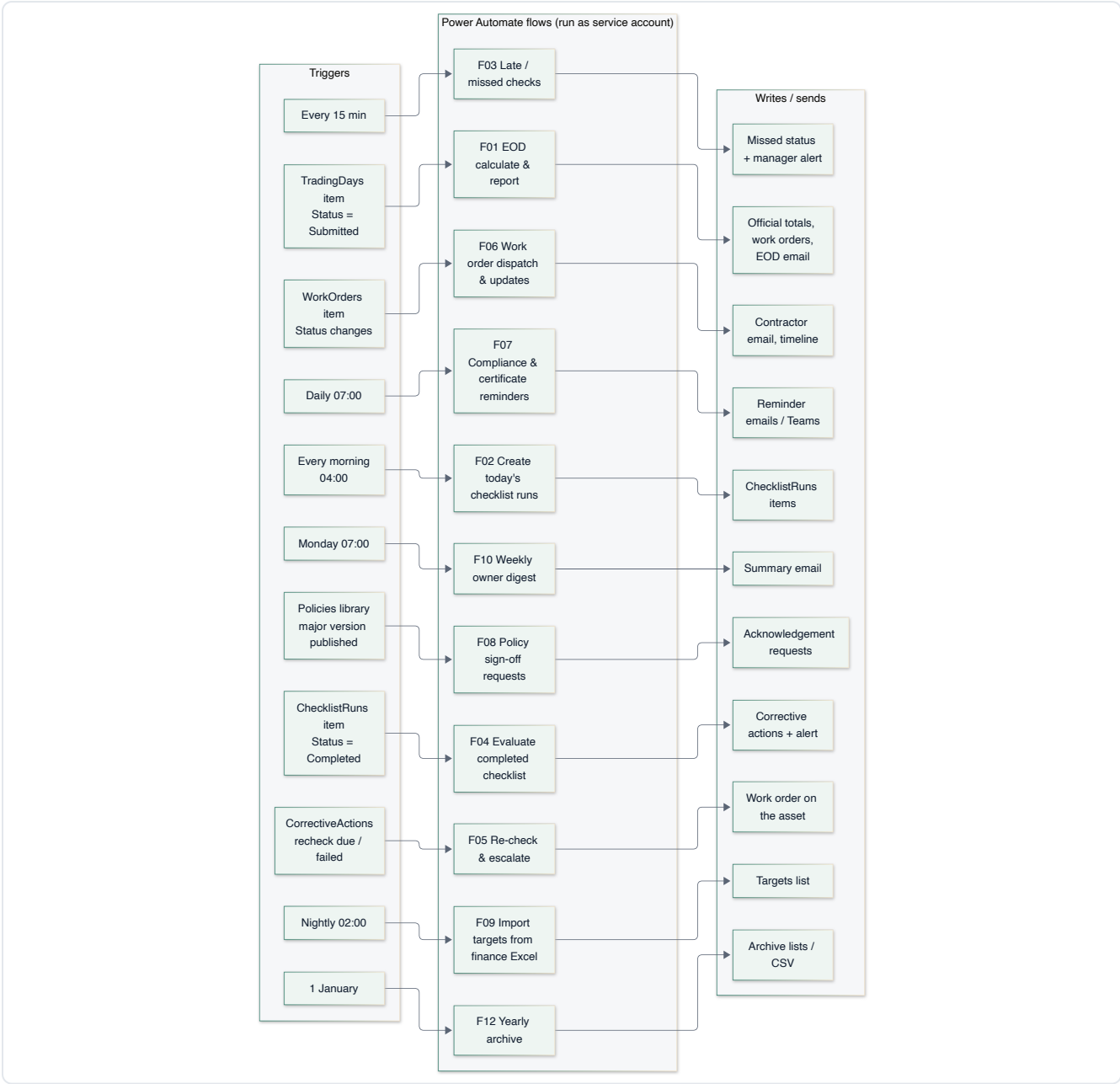
Library	Holds	Notes
Certificates	Staff and contractor certificates (gas CP42, TR19, EICR…)	Linked from StaffCertificates / ComplianceItems
Invoices	Contractor invoice PDFs	Finance + Admin only
Reports	Generated EOD, day-closure and EHO pack PDFs	Written by flows
Archive	Yearly CSV/list exports for archived years	Created by flow F12

Design conventions

- **Unique keys.** Rows that must exist only once use a composite key in `Title` with "enforce unique values" switched on, for example `BUN|2026-10-06` for a trading day, plus a `ClientId` for anything an offline device creates.
- **Filter columns.** Every large list has **indexed `Site` and `BusinessDate`** columns, and **every query filters on them**. Each list has at most 5 indexes including unique keys (SharePoint allows 20) and at most 5 lookup columns (a query allows 12).
- **Flow-owned columns.** Columns marked *set by flow* are written only by Power Automate, and the apps treat them as read-only.
- **Corrections.** A correction is a **new item** whose `Supersedes` column points to the old one, never an edit in place. Version history stays on.
- **Records labels.** Where Maginhawa's Microsoft 365 plan includes **Microsoft Purview retention labels** (check the licence), apply a "record" label to `Responses`, `CorrectiveActions` and submitted `TradingDays`. This blocks edits and deletes and makes the compliance record tamper-resistant.

Power Automate flows

All flows are **owned by a dedicated service account**, so they don't consume a manager's daily allowance or stop working when someone leaves. They use **standard connectors only** (SharePoint, Outlook, Teams, Excel Online) unless marked premium.



Flow	Trigger	What it does	Approx. runs per day (9 sites)
F01 EOD calculate & report	`TradingDays` item: Status → Submitted	Reads sessions, adjustments, delivery, targets and site settings. Calculates net (discounts subtracted), covers, ASPH, food/drink %, delivery estimated net, total net, wage % and RAG, and net vs target. Flags discount lines that don't match the EPOS total and cash variances with no explanation. Writes official totals; creates work orders from maintenance feedback; emails the EOD summary	~9
F02 Create today's checklist runs	Daily 04:00	Creates `ChecklistRuns` items from `Schedules`, skipping paused sites	1 (creates ~54 runs)
F03 Late / missed checks	Every 15 min during trading hours	Marks overdue runs Missed or Auto-closed and alerts the duty manager (Teams or email)	~60
F04 Evaluate completed checklist	`ChecklistRuns` item: Status → Completed	Re-checks that run's responses against `RulesThresholds`, sets `FailedCount`, creates corrective actions with re-check times and alerts the manager. **One run per checklist, not per answer**	~54
F05 Re-check & escalate	Every 15 min	Finds corrective actions past their re-check time with no passing re-check, or repeat failures on the same asset, and escalates them to a **work order** on that asset	~60 (most do nothing)
F06 Work order dispatch & updates	`WorkOrders` item: Status changes	Emails the contractor the job details (photos, asset, site contact), appends to the timeline and notifies the manager on resolution. Contractors reply by email or through a Microsoft Forms link (standard), so they need no login	~5–15

Flow	Trigger	What it does	Approx. runs per day (9 sites)
F07 Compliance & certificate reminders	Daily 07:00	Sends reminders for `ComplianceItems` due within 30 days, expiring `StaffCertificates` and expiring asset warranties	1
F08 Policy sign-off requests	Major version published in `Policies`	Notifies the affected roles and creates sign-off requests. A weekly chase covers anyone who hasn't signed	Occasional
F09 Import targets	Nightly 02:00	Reads finance's targets workbook (Excel Online connector) and creates or updates `Targets`	1
F10 Weekly owner digest	Monday 07:00	Summarises the week by site: sales vs target, wage %, missed checks, open repairs, expiring certificates	1
F12 Yearly archive	1 January	Copies last year's high-volume items to archive lists or CSV in the `Archive` library, then clears the live lists	1 per year

Finance's existing Flows 2 and 3 (the targets lookup and emails) are replaced by F01, F09 and F10. During the switch-over, F01 can also append the same row to finance's current Excel table, so nothing breaks while they move to the new views.

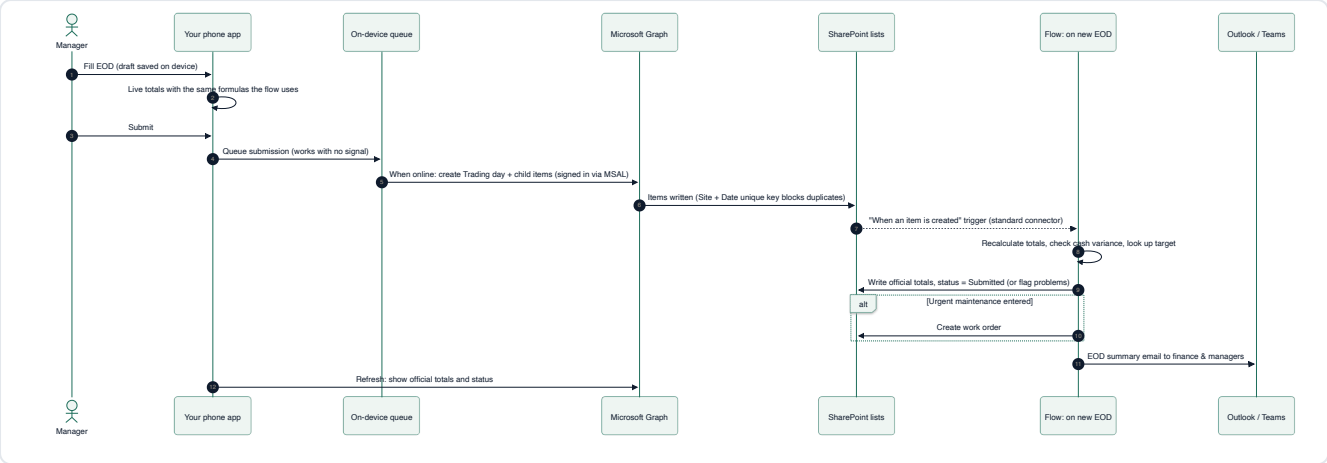
Trading / EOD module (replacing the current form)

Today's EOD form is a static Netlify page that calculates in the browser and posts JSON to a **public Power Automate URL**, which appends a row to Excel. A code review of it found five problems:

- The Power Automate signature is visible in the page source, so anyone could submit data.
- **Net sales ignores discounts and comps**, even though the form's label says it includes them.
- The delivery net (`gross ÷ 1.2 × 0.55`) and the wage target (35%) are hard-coded.
- Lunch managers are dropped from the submission.
- Discounts, comps, voids and refunds are flattened into text strings that can't be analysed.

The new module keeps the form's **23 discount/comp reasons**, **15 W/NW void codes** and staff positions as editable lists, and fixes the rest:

Area	Old form	New module
Identity	Typed names	Signed-in Microsoft user; approvers are real people
Calculations	Browser only; net omits discounts	App preview, then **flow F01** writes the official totals
Settings	20% VAT, 55% delivery retention, 35%/30% wage thresholds in code	Settings in the `Sites`, `DeliveryPlatforms` and `Targets` lists
Line items	Text strings	`Adjustments` rows with `ReasonCodes`, amount, approver and time
Cash	Warning only	Variance written by F01. The app requires an explanation above 1p, and F01 flags it if missing
Duplicates	Possible	Unique Site + Date key
Maintenance notes	Text in an email	A `Feedback` item that F01 turns into a **work order**
Excel	The only data store	A view that reads the lists (Power Query)



Do now, whatever else happens: regenerate the exposed Power Automate trigger URL.

AI in this architecture

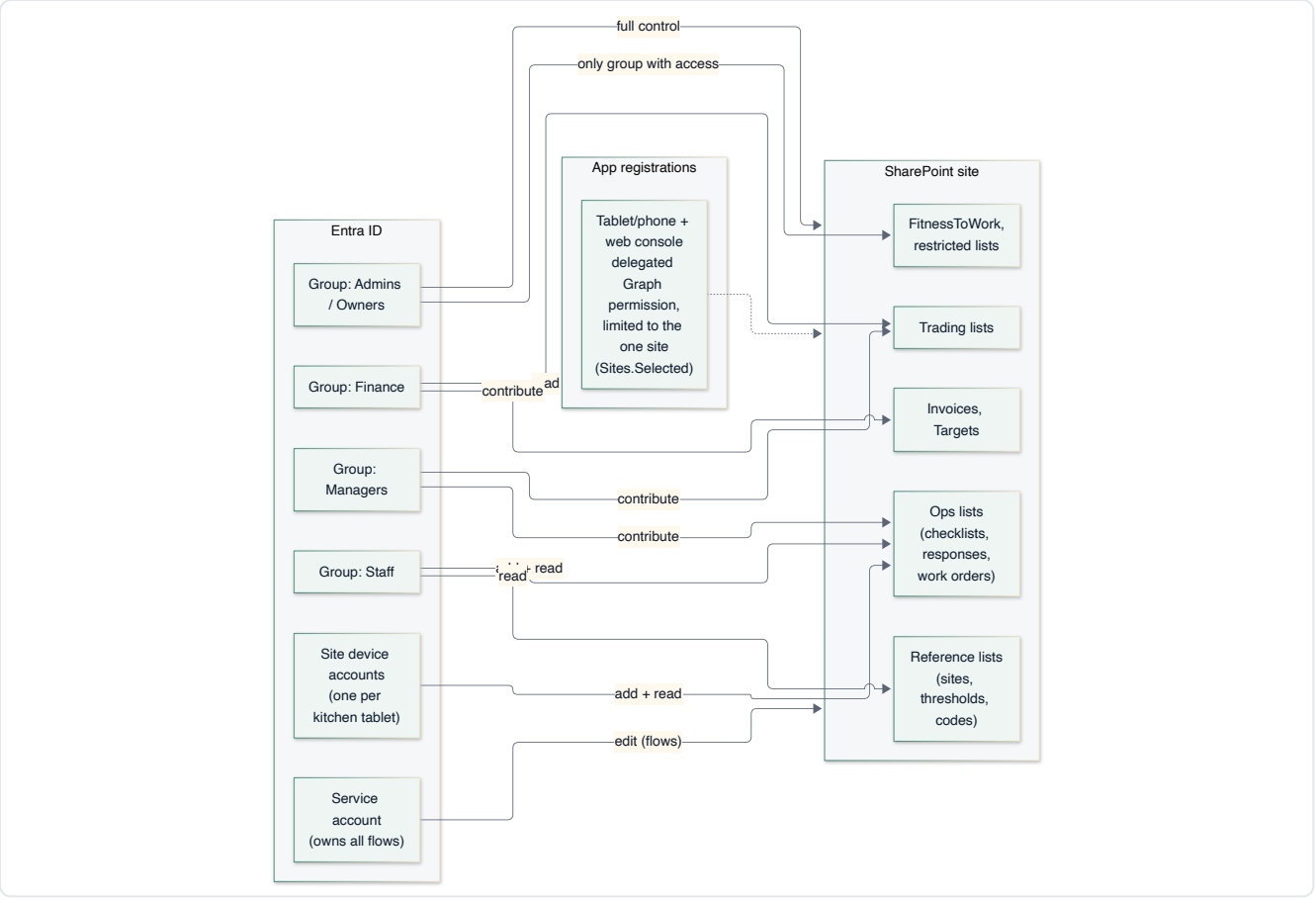
The apps **cannot hold an AI API key**, because anyone could extract it from the app. So AI arrives in Stage 3 with the **first small piece of our own backend**: an **Azure Function "AI proxy"** in the UK South region. It holds the Claude API key, checks the user's Microsoft sign-in token and queries SharePoint through Graph **as that user**, so it can never see more than the person asking. That enables the features from the product plan:

- **Ask your logs**, with tool calls and record citations
- **Ask the handbook**
- **Reading thermometer photos and invoices**
- **AI-written EOD summaries**

Until then, **Copilot in Power BI** (it needs a specific paid capacity, so check) can answer natural-language questions over the finance dataset.

Security and data protection

- **Data stays in Maginhawa's Microsoft 365**, in its existing data location, so no new processor is needed for core data. The AI proxy, when added, means signing Anthropic's data processing terms and keeping to UK GDPR data minimisation. Never send health or HR lists to the AI.
- **App registrations** use **delegated Graph permissions limited to the one SharePoint site** (`Sites.Selected`), so the apps can't read anything else in the organisation.
- **No public trigger URLs.** Flows start from list changes or schedules.
- **Permissions by group.** Staff and device accounts can add and read operational lists. Finance has contribute rights on finance lists and read rights on trading lists. Restricted lists (`FitnessToWork`) are open only to Owners/Admins. No item-level permissions are used, because they don't scale.
- **Device management.** Use Conditional Access and Intune where licensed: managed devices only for device accounts, MFA for managers and finance, and remote wipe or revocation (`Devices.Revoked`).
- **Audit.** Version history and SharePoint audit logs, plus optional Purview record labels for tamper-resistance.



Scalability and limits

Microsoft limits change, so **check the current documentation** before relying on any number below.

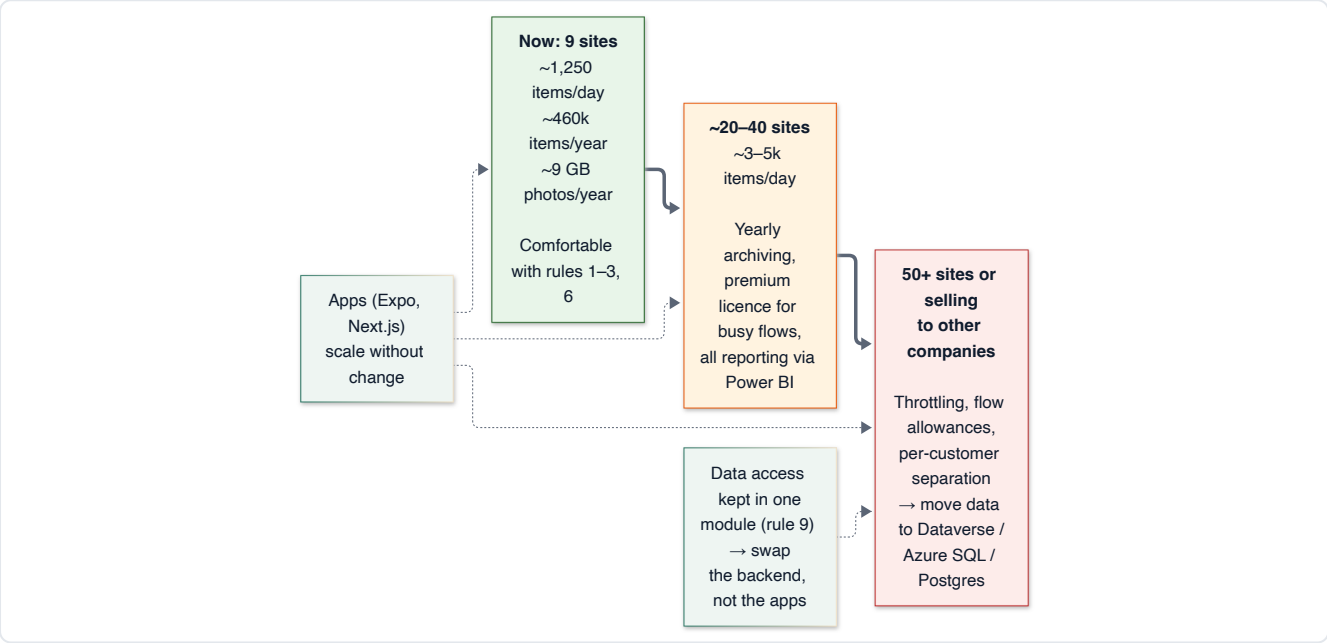
Expected volume at Maginhawa (9 sites, estimate): about 1,250 new list items a day (≈800 checklist answers plus runs, EOD rows, discounts/comps/voids and staff rows). That is **about 460k items a year**, plus about 80 photos a day, **about 9 GB a year** once compressed.

Component	Limit	Risk at 9 sites
Our apps (Expo, Next.js on Netlify)	Effectively none	✅ None
SharePoint list size	~30 million items per list	✅ Fine
5,000-item list view threshold	Queries that don't filter on an **indexed** column fail once a list has more than 5,000 items; up to 20 indexes per list	⚠️ `Responses` passes 5,000 within weeks. Index and always filter (rule 2)
Lookup columns per query	12	✅ Max 5 per list in this design
Microsoft Graph throttling	Per-user and per-app limits; returns 429 + `Retry-After`	✅ Low; the app retries (rule 7)
Power Automate daily action allowance	Per licence, **charged to the flow owner**; standard Microsoft 365 allowances are modest	⚠️ **Main risk.** A flow per checklist answer would use thousands of actions a day (rule 1)
Flow trigger delay	SharePoint triggers poll, so a flow can take minutes to start on standard licences	⚠️ Fine for EOD and alerts. The app shows an instant preview, so users never wait
Storage	Organisation pool ~1 TB plus extra per licence	✅ Fine with on-device compression (rule 5)
Reporting across history	SharePoint isn't an analytics engine	⚠️ Power BI imports on a schedule (rule 8)

Rules that keep it scaling

1. **Don't run a flow per checkpoint answer.** Trigger on the **completed checklist** (F04), or run a scheduled batch every 10–15 minutes. This is the biggest saving.

2. **Index `Site`, `BusinessDate` and `Status`, and always filter on them.** Never load a whole list, in the app, the flows or Power Query.
3. **Archive yearly** (F12). Keep the live lists to the current year.
4. **Batch writes** with Graph `$batch` (20 per call) and **cache reference data** on the device.
5. **Compress photos on the device** before upload.
6. **Flows are owned by a service account** with a licence suited to the load. Move the busiest flows to a premium or per-flow licence if allowances run short.
7. **Honour `Retry-After`** when throttled. The offline queue does this automatically.
8. **Power BI uses scheduled imports**, not live list queries.
9. **Keep all data access in one module of our code.** Swapping SharePoint for Dataverse, Azure SQL or Postgres then only touches that module.

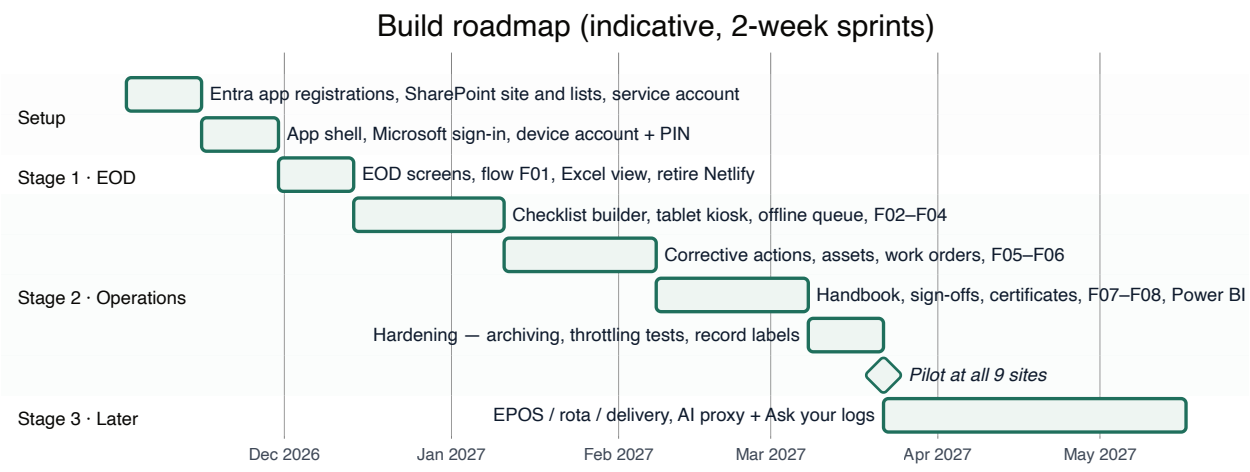


When to move off SharePoint: at **50+ sites**, when **selling to other companies**, when you need **true record immutability** without Purview, when you need **heavy AI or analytics over history**, or when flow allowances keep running out despite rules 1 and 6. Thanks to rule 9, the apps don't change.

Licensing checklist

Item	Standard (in Microsoft 365)	Premium / extra
SharePoint, Outlook, Teams, Excel Online connectors	✅	
Graph API from our apps	✅	
Flows triggered by list changes or schedules	✅	Busy flows may need a premium/process licence (rule 6)
HTTP-trigger flows, SMS (Twilio), Word→PDF templates, Dataverse, custom connectors		£ Premium
Power BI viewers		£ Power BI Pro (unless already licensed)
Purview record labels, Intune, Conditional Access	Depends on plan (e.g. Business Premium / E3)	Check
Kitchen tablet device accounts		Confirm with the Microsoft partner
Azure Function AI proxy (Stage 3)		£ Small Azure consumption + Claude API usage

Build order



Sprint (2 weeks)	Build
1	Entra app registrations (`Sites.Selected`), service account, **SharePoint site and lists provisioned from `sharepoint-lists.json`** , seed reference data (sites, reason codes, positions, platforms)
2	App shell (Expo + Next.js monorepo), MSAL sign-in, device account + PIN, the `data` module over Graph with an offline queue
3	**EOD screens + F01 + Excel view; retire the Netlify form** (Stage 1 quick win)
4–5	Checklist builder (console), tablet kiosk checklists with media, F02, F03, F04
6–7	Corrective actions, assets + QR, work orders, contractors, F05, F06, day closure and spreadsheet view
8–9	Handbook and policies, sign-offs, certificates, F07, F08, F09, F10; Power BI dashboards
10	Hardening: indexes check, load test at 3× volume, throttling test, F12 archive dry run, Purview labels; **pilot at all 9 sites**
Stage 3	EPOS, rota and delivery auto-fill; AI proxy + Ask your logs; statutory calendar, allergens, probes

Decisions to confirm

1. **Microsoft 365 plan** (Business Standard, Business Premium or E3/E5). This decides Purview labels, Intune and Conditional Access.
2. **Device accounts for kitchen tablets.** Confirm the licensing approach with the Microsoft partner.
3. **Power BI licences** for finance and owners.
4. **Net sales formula.** Is EPOS "gross" before or after discounts? This is set by `Sites.NetSalesRule`.
5. **Delivery estimate.** Keep 55% retention per platform, or capture actual payouts weekly?
6. **Finance's Excel.** Keep the existing workbook layout, fed by F01 during the switch-over, or move straight to the Power Query view?
7. **Who owns the service account and the flows** inside Maginhawa's IT.
8. **Food-safety thresholds and inspection intervals.** Every `RulesThresholds` and `ComplianceItems` value starts unverified until a consultant signs it off.

Appendix A: SharePoint list specification

Tags: unique = "enforce unique values" (indexed); indexed = column index; set by flow = written only by Power Automate.

Reference

Sites: One row per restaurant; drives rules, targets and settings. *Written by:* Admin (console).

Column	Type	Notes
`Title` *(required)*	Text	Restaurant name, e.g. Bunso
`SiteCode` *(unique, required)*	Text	Short code used in keys, e.g. BUN
`Nation` *(required)*	Choice	England / Wales / Scotland / Northern Ireland (selects rules pack)
`Timezone`	Text	Europe/London
`FHRSId`	Number	FSA establishment id for rating sync
`Sessions`	Choice (multi)	Lunch, Dinner, All day
`VATRatePct`	Number	Default 20
`WageTargetPct`	Number	Red above (default 35)
`WageAmberPct`	Number	Amber above (default 30)
`NetSalesRule`	Choice	GrossMinusSCTaxDiscounts (default) / GrossMinusSCTax — confirm with finance
`DayCloseTime`	Text	e.g. 23:30 — when EOD/missed-check jobs run
`Active` *(indexed)*	Yes/No	

Indexed (2): SiteCode, Active

Staff: Everyone who uses the apps, incl. kiosk-only staff without Microsoft accounts. *Written by:* Admin / Manager.

Column	Type	Notes
`Title` *(required)*	Text	Full name
`Email` *(indexed)*	Text	Microsoft account (blank for kiosk-only staff)
`HomeSite` *(indexed)*	Lookup → Sites	
`Department`	Choice	Kitchen / FOH / Management / Finance / Admin
`Position`	Choice	Head chef, Sous chef, CDP, Commis, KP, Floor manager, Senior waiter, Waiter, Runner, Host, Bartender, RM, GM, AM, Duty manager, Supervisor
`AppRole` *(required)*	Choice	Staff / Manager / Admin / Finance / Owner
`PinHash`	Text	argon2 hash of 4-digit kiosk PIN (never the PIN)
`PreferredLanguage`	Text	en-GB default
`Active` *(indexed)*	Yes/No	

Indexed (3): Email, HomeSite, Active

Devices: Registered kiosk tablets and phones. *Written by:* Admin.

Column	Type	Notes
`Title` *(required)*	Text	Label, e.g. Bunso kitchen tablet
`Site` *(indexed)*	Lookup → Sites	
`DeviceAccount`	Text	Site device Microsoft account (UPN)
`Platform`	Choice	iPadOS / Android / iOS / Web
`LastSeen`	Date/Time	Updated by app
`Revoked` *(indexed)*	Yes/No	Revoked devices are refused by the app

Indexed (2): Site, Revoked

Targets: Daily targets per site (imported from finance Excel or edited in console). *Written by:* Flow F09 / Finance.

Column	Type	Notes
`Title` *(unique, required)*	Text	Key = SiteCode|yyyy-mm-dd
`Site` *(indexed)*	Lookup → Sites	
`TargetDate` *(indexed)*	Date	
`NetSalesTarget`	Currency	
`GrossSalesTarget`	Currency	
`CoversTarget`	Number	
`WagePctTarget`	Number	Overrides site default

Indexed (3): Title, Site, TargetDate

ReasonCodes: Discount / comp / void reasons (23 discount/comp + 15 void codes from the current form). *Written by:* Finance / Admin.

Column	Type	Notes
`Title` *(required)*	Text	Label, e.g. Employee Dining up to 4 Food Only
`Kind` *(indexed, required)*	Choice	Discount / Comp / Void
`Scope`	Choice	Check / Item
`DefaultPct`	Number	20 / 50 / 100; blank = chosen amount
`IsWaste`	Yes/No	Voids: W (waste) vs NW
`AppliesTo`	Choice	Food / Drinks / Food & drink
`NotePrompt`	Text	What the note must contain
`ApprovalRole`	Choice	Manager / Owner
`SortOrder`	Number	
`Active` *(indexed)*	Yes/No	

Indexed (2): Kind, Active

DeliveryPlatforms: Deliveroo, Uber Eats, Just Eat…. *Written by:* Finance.

Column	Type	Notes
`Title` *(unique, required)*	Text	Platform name
`EstimatedNetPct`	Number	Replaces hard-coded 55% (net ≈ gross ÷ (1+VAT) × pct)
`PricesIncludeVAT`	Yes/No	
`Active`	Yes/No	

Indexed (1): Title

RulesThresholds: Food-safety limits per nation (data, not code); consultant-verified. *Written by:* Admin (consultant sign-off).

Column	Type	Notes
`Title` *(unique, required)*	Text	Key = Nation|CheckCode
`Nation` *(indexed)*	Choice	
`CheckCode`	Text	cold_hold, hot_hold, reheat, cook_core, freezer, cooling_stage1 …
`MinValue`	Number	
`MaxValue`	Number	
`Unit`	Text	C
`LegalBasis`	Text	Citation shown to EHO
`Verified`	Yes/No	False until checked against primary source

Indexed (2): Title, Nation

Checklists

ChecklistTemplates: Checklist definitions (opening, closing, cleaning, food safety, custom). *Written by:* Admin / Manager (console builder).

Column	Type	Notes
`Title` *(required)*	Text	Name
`Kind` *(indexed)*	Choice	Opening / Closing / Cleaning / Food safety / Delivery / Handover / Custom
`CurrentVersion`	Number	Bumped on publish; runs store the version used
`Status` *(indexed)*	Choice	Draft / Published / Archived
`Sites`	Lookup (multi) → Sites	Where it applies

Indexed (2): Kind, Status

Checkpoints: Questions inside a template version — the customisable checkpoints with media. *Written by:* Admin / Manager (builder).

Column	Type	Notes
`Title` *(required)*	Text	Prompt, e.g. Walk-in chiller temperature
`Template` *(indexed)*	Lookup → ChecklistTemplates	
`TemplateVersion` *(indexed)*	Number	Published versions are never edited
`Position`	Number	
`Section`	Text	
`Guidance`	Multi-line text	'How to do it right' description
`MediaLinks`	Multi-line text	Links to images/videos in HowToMedia library
`ResponseType` *(required)*	Choice	Yes/No, Number, Temperature, Text, Select, Multi-select, Photo, Signature, Date, Time
`Options`	Multi-line text	JSON for select options
`CheckCode`	Text	Links to RulesThresholds for pass/fail
`Asset`	Lookup → Assets	Optional: checkpoint about a specific unit
`Required`	Yes/No	
`PhotoRequired`	Yes/No	
`CorrectiveActionOnFail`	Yes/No	

Indexed (2): Template, TemplateVersion

Schedules: When each checklist is due at each site. *Written by:* Admin / Manager.

Column	Type	Notes
`Title`	Text	Description
`Template` *(indexed)*	Lookup → ChecklistTemplates	
`Site` *(indexed)*	Lookup → Sites	
`DueTime` *(required)*	Text	e.g. 08:00
`Days`	Choice (multi)	Mon–Sun
`WindowMinutes`	Number	Late after this
`AutoClose`	Yes/No	
`AssignedRole`	Choice	

Column	Type	Notes
`PausedFrom`	Date	Closures / refurb
`PausedUntil`	Date	

Indexed (2): Template, Site

ChecklistRuns: One instance of a checklist on a day (created each morning by flow F02). *Written by:* Flow F02 creates; app completes.

Column	Type	Notes
`Title` *(unique, required)*	Text	Key = SiteCode|yyyy-mm-dd|ScheduleId
`Site` *(indexed)*	Lookup → Sites	
`BusinessDate` *(indexed)*	Date	
`Template`	Lookup → ChecklistTemplates	
`TemplateVersion`	Number	
`DueAt`	Date/Time	
`Status` *(indexed)*	Choice	Scheduled / In progress / Completed / Missed / Auto-closed / Cancelled
`CompletedBy`	Lookup → Staff	
`CompletedAt`	Date/Time	
`DurationSeconds`	Number	Pencil-whip signal
`FailedCount` *(set by flow)*	Number	Set by flow F04

Indexed (4): Title, Site, BusinessDate, Status

Responses: Every answer — the compliance record (largest list). *Written by:* App (append-only by convention + Purview record label).

Column	Type	Notes
`Title` *(unique, required)*	Text	ClientId — UUID from the device; makes offline retries idempotent
`Run` *(indexed)*	Lookup → ChecklistRuns	
`Site` *(indexed)*	Lookup → Sites	
`BusinessDate` *(indexed)*	Date	
`CheckpointId`	Number	Checkpoint item id
`ValueNumber`	Number	
`ValueText`	Multi-line text	Text / select / signature JSON
`Passed` *(indexed)*	Yes/No	App preview; confirmed by flow F04
`ThresholdSnapshot`	Text	Min/max/legal basis used, frozen
`Source`	Choice	Manual / Bluetooth probe / Camera OCR
`PhotoUrl`	Hyperlink	Photos library
`PhotoSha256`	Text	Detects reused photos
`CapturedLive`	Yes/No	In-app camera, not gallery
`RecordedBy`	Lookup → Staff	
`Device`	Lookup → Devices	
`ClientRecordedAt`	Date/Time	Device clock (Created = server time)
`Supersedes`	Number	Id of the response this corrects (never edit in place)
`CorrectionReason`	Text	

Indexed (5): Title, Run, Site, BusinessDate, Passed

CorrectiveActions: What was done about a failed check. *Written by:* App + flows F04/F05.

Column	Type	Notes
`Title` *(required)*	Text	Short description
`Response` *(indexed)*	Lookup → Responses	
`Site` *(indexed)*	Lookup → Sites	
`ActionTaken`	Multi-line text	
`FoodDisposed`	Yes/No	
`Status` *(indexed)*	Choice	Open / Action taken / Verified / Escalated
`RecheckDue` *(indexed)*	Date/Time	
`WorkOrder`	Lookup → WorkOrders	Set when escalated
`VerifiedBy`	Lookup → Staff	

Indexed (4): Response, Site, Status, RecheckDue

DayClosures: Ops end-of-day sign-off per site. *Written by:* App + flow F01.

Column	Type	Notes
`Title` *(unique, required)*	Text	Key = SiteCode|yyyy-mm-dd
`Site` *(indexed)*	Lookup → Sites	
`BusinessDate` *(indexed)*	Date	
`ClosedBy`	Lookup → Staff	Blank = auto-closed
`Stats` *(set by flow)*	Multi-line text	JSON: completion %, failures, open CAs, faults
`ReportLink` *(set by flow)*	Hyperlink	PDF in Reports library

Indexed (3): Title, Site, BusinessDate

HandoverNotes: Shift handover log. *Written by:* App.

Column	Type	Notes
`Title`	Text	Summary
`Site` *(indexed)*	Lookup → Sites	
`BusinessDate` *(indexed)*	Date	
`Body`	Multi-line text	
`Author`	Lookup → Staff	

Indexed (2): Site, BusinessDate

Maintenance

Assets: Equipment register with QR sticker per unit. *Written by:* Manager / Admin.

Column	Type	Notes
`Title` *(required)*	Text	e.g. Walk-in chiller
`Site` *(indexed)*	Lookup → Sites	
`Category` *(indexed)*	Choice	Refrigeration, Cooking, Extraction, Gas, Electrical, Dishwash, Ice, Coffee, Lift, Other
`Make`	Text	
`Model`	Text	
`SerialNumber`	Text	
`QRCode` *(unique)*	Text	Value on the sticker
`InstallDate`	Date	
`WarrantyExpires` *(indexed)*	Date	

Column	Type	Notes
`DefaultContractor`	Lookup → Contractors	
`PhotoUrl`	Hyperlink	
`Retired`	Yes/No	

Indexed (4): Site, Category, QRCode, WarrantyExpires

Contractors: Refrigeration, gas, electrical, extraction, pest…. *Written by:* Manager / Admin.

Column	Type	Notes
`Title` *(required)*	Text	Company
`Trade` *(indexed)*	Choice	
`ContactName`	Text	
`Phone`	Text	
`Email`	Text	
`GasSafeNumber`	Text	
`Accreditation`	Text	e.g. BESA TR19

Indexed (1): Trade

WorkOrders: Repairs: from fault report, failed check, statutory due or EOD note. *Written by:* App + flows F05/F06.

Column	Type	Notes
`Title` *(unique, required)*	Text	Reference WO-2026-00123
`Site` *(indexed)*	Lookup → Sites	
`Asset` *(indexed)*	Lookup → Assets	
`Summary` *(required)*	Text	
`Description`	Multi-line text	
`Source`	Choice	Fault report / Failed check / Statutory / Planned / EOD note / AI warning
`SourceResponse`	Number	Response id when source = failed check
`Priority` *(indexed)*	Choice	Low / Normal / High / Urgent
`Status` *(indexed)*	Choice	Reported / Triaged / Dispatched / Scheduled / In progress / Awaiting parts / Resolved / Closed / Cancelled
`Contractor`	Lookup → Contractors	
`ReportedBy`	Lookup → Staff	
`DispatchedAt`	Date/Time	
`AttendedAt`	Date/Time	
`ResolvedAt`	Date/Time	
`PhotoUrls`	Multi-line text	
`History`	Multi-line text (append)	Timeline notes (append changes on)

Indexed (5): Title, Site, Asset, Priority, Status

ComplianceItems: Statutory calendar: gas CP42, TR19, EICR, fire alarm, emergency lighting, LOLER, PSSR, legionella, pest. *Written by:* Admin / flow F07.

Column	Type	Notes
`Title` *(required)*	Text	Requirement name
`Site` *(indexed)*	Lookup → Sites	

Column	Type	Notes
`Asset`	Lookup → Assets	Optional
`RequirementCode`	Choice	
`IntervalMonths`	Number	[verify] per requirement
`LastDone`	Date	
`NextDue` *(indexed)*	Date	
`Contractor`	Lookup → Contractors	
`CertificateLink`	Hyperlink	Certificates library
`Verified`	Yes/No	Interval checked by consultant

Indexed (2): Site, NextDue

Invoices: Contractor invoices for the Finance role. *Written by:* Finance.

Column	Type	Notes
`Title` *(required)*	Text	Invoice number
`Contractor` *(indexed)*	Lookup → Contractors	
`WorkOrder` *(indexed)*	Lookup → WorkOrders	
`Asset` *(indexed)*	Lookup → Assets	
`InvoiceDate` *(indexed)*	Date	
`Net`	Currency	
`VAT`	Currency	
`Gross`	Currency	
`DocumentLink`	Hyperlink	Invoices library
`ApprovedBy`	Lookup → Staff	
`ExportedAt`	Date/Time	To Xero / QuickBooks / Sage (phase 2)

Indexed (4): Contractor, WorkOrder, Asset, InvoiceDate

VisitorLog: Contractor and visitor sign-in. *Written by:* App.

Column	Type	Notes
`Title` *(required)*	Text	Visitor name
`Site` *(indexed)*	Lookup → Sites	
`Company`	Text	
`Purpose`	Text	
`WorkOrder`	Lookup → WorkOrders	
`SignedIn` *(indexed)*	Date/Time	
`SignedOut`	Date/Time	
`HealthDeclaration`	Yes/No	Food-handler fitness questions answered

Indexed (2): Site, SignedIn

Trading (EOD)

TradingDays: One per site per service date — replaces the Netlify form's Excel row. *Written by:* App creates; flow F01 writes official totals.

Column	Type	Notes
`Title` *(unique, required)*	Text	Key = SiteCode|yyyy-mm-dd (blocks duplicates)
`Site` *(indexed)*	Lookup → Sites	

Column	Type	Notes
`BusinessDate` *(indexed)*	Date	
`Status` *(indexed)*	Choice	Draft / Submitted / Checked / Approved / Reopened
`WageCost`	Currency	
`NextShiftNotes`	Multi-line text	
`SubmittedBy`	Lookup → Staff	Signed-in user, not typed
`SubmittedAt`	Date/Time	
`ApprovedBy`	Lookup → Staff	Finance sign-off
`TotalGross` *(set by flow)*	Currency	
`TotalNet` *(set by flow)*	Currency	
`DineInNet` *(set by flow)*	Currency	
`Covers` *(set by flow)*	Number	
`ASPHNet` *(set by flow)*	Currency	
`FoodPct` *(set by flow)*	Number	
`DrinkPct` *(set by flow)*	Number	
`WagePct` *(set by flow)*	Number	
`WageRAG` *(set by flow)*	Choice	Green / Amber / Red
`NetVsTarget` *(set by flow)*	Currency	
`CheckIssues` *(set by flow)*	Multi-line text	Problems found by flow (e.g. discount lines ≠ EPOS total)
`CalcVersion` *(set by flow)*	Text	Formula version used

Indexed (4): Title, Site, BusinessDate, Status

TradingSessions: Lunch / dinner figures and cash. *Written by:* App; flow F01 adds net & variance.

Column	Type	Notes
`Title` *(unique, required)*	Text	Key = SiteCode|yyyy-mm-dd|Session
`TradingDay` *(indexed)*	Lookup → TradingDays	
`Site` *(indexed)*	Lookup → Sites	
`BusinessDate` *(indexed)*	Date	
`Session`	Choice	Lunch / Dinner / All day
`Gross`	Currency	
`ServiceCharge`	Currency	
`Tax`	Currency	
`DiscountsComps`	Currency	EPOS total; reconciled to Adjustments
`FoodSales`	Currency	
`DrinkSales`	Currency	
`BookedCovers`	Number	'Starting covers' in old form
`Walkins`	Number	
`CashInEpos`	Currency	
`CashCollected`	Currency	
`CashExplanation`	Multi-line text	Required by app if variance > 1p
`Net` *(set by flow)*	Currency	Gross – SC – tax – discounts
`Covers` *(set by flow)*	Number	
`CashVariance` *(set by flow)*	Currency	Collected – EPOS

Indexed (4): Title, TradingDay, Site, BusinessDate

Adjustments: Discounts, comps and voids as rows (were text blobs). *Written by:* App.

Column	Type	Notes
`Title` *(unique, required)*	Text	ClientId (UUID)
`TradingSession` *(indexed)*	Lookup → TradingSessions	
`Site` *(indexed)*	Lookup → Sites	
`BusinessDate` *(indexed)*	Date	
`Kind` *(indexed)*	Choice	Discount / Comp / Void
`ReasonCode` *(required)*	Lookup → ReasonCodes	
`GuestOrStaffName`	Text	
`Items`	Text	
`SaleAmount`	Currency	
`Amount` *(required)*	Currency	
`Time`	Text	
`PerformedBy`	Lookup → Staff	'Voided by'
`ApprovedBy`	Lookup → Staff	Real user
`Notes`	Multi-line text	

Indexed (5): Title, TradingSession, Site, BusinessDate, Kind

DeliverySales: Per platform per day. *Written by:* App; flow F01 adds estimated net.

Column	Type	Notes
`Title` *(unique, required)*	Text	Key = SiteCode|yyyy-mm-dd|Platform
`TradingDay` *(indexed)*	Lookup → TradingDays	
`Platform`	Lookup → DeliveryPlatforms	
`Gross`	Currency	
`EstimatedNet` *(set by flow)*	Currency	
`ActualPayout`	Currency	From platform statement (later)
`CurrentPromotion`	Text	

Indexed (2): Title, TradingDay

DeliveryRefunds: Refund requests from delivery platforms. *Written by:* App.

Column	Type	Notes
`Title`	Text	Summary
`TradingDay` *(indexed)*	Lookup → TradingDays	
`Platform`	Lookup → DeliveryPlatforms	
`Amount` *(required)*	Currency	
`Details`	Multi-line text	
`Issued` *(indexed)*	Choice	Yes / No / Pending

Indexed (2): TradingDay, Issued

ShiftStaff: Who worked each session (fixes lunch managers being dropped). *Written by:* App (rota later).

Column	Type	Notes
`Title` *(required)*	Text	Name
`TradingSession` *(indexed)*	Lookup → TradingSessions	
`Staff`	Lookup → Staff	Blank if no account
`Department`	Choice	Kitchen / FOH / Management
`Position`	Choice	

Column	Type	Notes
`Hours`	Number	

Indexed (1): TradingSession

Feedback: Service, kitchen, staff, complaints, maintenance notes. *Written by:* App; flow F01 links work orders.

Column	Type	Notes
`Title`	Text	Category
`TradingDay` *(indexed)*	Lookup → TradingDays	
`Category` *(indexed)*	Choice	Service / Kitchen / Staff / Complaints / Maintenance
`Body` *(required)*	Multi-line text	
`WorkOrder` *(set by flow)*	Lookup → WorkOrders	Created by F01 for urgent maintenance

Indexed (2): TradingDay, Category

People & handbook

PolicyAcknowledgements: E-signed policy sign-offs. *Written by:* App.

Column	Type	Notes
`Title` *(unique, required)*	Text	Key = PolicyFileId|Version|StaffId
`PolicyFileId` *(indexed)*	Number	Item id in Policies library
`PolicyVersion`	Text	Library version label at signing
`Staff` *(indexed)*	Lookup → Staff	
`AcknowledgedAt`	Date/Time	
`Device`	Lookup → Devices	
`SignatureJson`	Multi-line text	

Indexed (3): Title, PolicyFileId, Staff

StaffCertificates: Level 2/3 food hygiene, personal licence, first aid…. *Written by:* Manager.

Column	Type	Notes
`Title` *(required)*	Text	Certificate type
`Staff` *(indexed)*	Lookup → Staff	
`Issuer`	Text	
`IssuedOn`	Date	
`ExpiresOn` *(indexed)*	Date	
`DocumentLink`	Hyperlink	Certificates library

Indexed (2): Staff, ExpiresOn

FitnessToWork: Illness exclusion / return to work — restricted to Owner/Admin. *Written by:* Manager (restricted list).

Column	Type	Notes
`Title` *(required)*	Text	Staff name
`Staff` *(indexed)*	Lookup → Staff	
`ReportedOn`	Date	
`SymptomsCategory`	Choice	Minimal data — no free-text diagnosis
`ExcludedUntil`	Date	

Column	Type	Notes
`ClearedBy`	Lookup → Staff	
`ClearedOn`	Date	

Indexed (1): Staff

Appendix B: the database option (if we ever sell it)

The original architecture uses **Postgres on Supabase** in London, PowerSync offline sync and a small Node API. Its full write-up, tested `schema.sql` (67 tables) and diagrams are kept in `postgres-option/`. Use it if the product is sold to other restaurants or grows beyond ~50 sites. The SharePoint lists mirror those tables, so migrating is a data copy, and because of rule 9 the apps keep working.